

Data Analysis and Visualization Prompt Collection

Professional prompt templates with refinement prompts in colored boxes

Core Analysis Prompts

1. Data Exploration Prompt

Prompt

You are a data analyst.

Analyze the dataset I provide and identify:

- Key trends
- Outliers
- Correlations between variables
- Unexpected patterns

Then summarize the insights in clear bullet points suitable for a professional report. Also suggest the best visualizations to represent each insight.

2. Professional Dashboard Prompt

Prompt

Act as a senior data analyst.

Using the dataset below, design a professional dashboard. Include:

- The 5 most important KPIs
- Recommended charts (bar, line, heatmap, scatter, etc.)
- Short explanations for each visualization
- Insights a decision-maker should focus on

Explain the reasoning behind the chosen visualizations.

3. Visualization Generation Prompt

Prompt

Analyze the dataset and create clear, professional visualizations.

For each visualization include:

- Title
- Type of chart
- What the chart shows
- Key insight

Focus on visuals that communicate trends quickly to executives. Avoid unnecessary charts.

4. Storytelling with Data Prompt

Prompt

Act as a data storyteller.

Analyze the dataset and produce:

1. Key insights
2. Visualizations that highlight the most important findings
3. A short narrative explaining what the data means and why it matters

The goal is to communicate impact clearly to non-technical stakeholders.

5. Advanced Analytics Prompt

Prompt

Act as a senior data scientist.

Analyze the dataset and perform:

- Trend analysis
- Correlation analysis
- Segmentation or clustering (if applicable)
- Predictive insights

Recommend visualizations that best communicate the results. Explain any important patterns discovered.

6. Quick Insight Prompt

Prompt

Analyze this dataset quickly.

Provide:

- Top 3 insights
- Top 3 recommended visualizations
- One surprising or unexpected finding

Refinement Prompts for Better Visualizations

1. Improve Readability

Prompt

Refine the visualization to maximize readability.

Adjust:

- Font sizes for titles, labels, and annotations
- Spacing between elements
- Clarity of axis labels
- Legend placement

Ensure the chart is easily readable when viewed in a presentation slide.

2. Optimize Scales and Axes

Prompt

Review the visualization and improve axis scaling.

Check:

- Whether the axis range accurately represents the data
- If logarithmic or normalized scales would be clearer
- Tick mark spacing and labeling
- Removal of unnecessary decimals

Explain any scale adjustments made.

3. Improve Color Palette

Prompt

Refine the visualization using a professional color palette.

Requirements:

- Ensure strong contrast for readability
- Avoid overly saturated colors
- Use colorblind-safe palettes
- Highlight the most important variable with a stronger color

Explain why the chosen palette improves the chart.

4. Enhance Visual Hierarchy

Prompt

Improve the visual hierarchy of the chart.

Focus on:

- Making the main insight visually dominant
- Reducing visual clutter
- Emphasizing key data points
- Adjusting line thickness, marker size, and label prominence

5. Improve Annotations and Labels

Prompt

Refine the chart annotations.

Add or improve:

- Descriptive title
- Concise subtitle explaining context
- Labels for important data points
- Short annotations highlighting key insights

Avoid excessive text.

6. Professional Presentation Style

Prompt

Refine the visualization for a professional report or executive presentation.

Improve:

- Layout balance
- Alignment of elements
- Consistent typography
- Clean background and gridlines
- Removal of unnecessary decorative elements

7. Dashboard Refinement

Prompt

Refine the dashboard layout.

Improve:

- Spacing between charts
- Consistent color scheme across visuals
- Clear grouping of related metrics
- Readable KPI cards
- Logical visual flow for viewers

8. Accessibility Refinement

Prompt

Improve accessibility of the visualization.

Ensure:

- Colorblind-friendly palette
- Sufficient contrast ratios
- Readable fonts
- Labels that do not rely solely on color

9. Meta-Refinement Prompt

Prompt

Critically review this visualization as if you were a data visualization expert.

Identify weaknesses in:

- Readability
- Scaling
- Color usage
- Labeling
- Clarity of the main insight

Then suggest precise improvements.